

MATH3171 (Mathematical Biology III)
YEAR 2025–26

PAST PAPER GUIDE

This guide suggests how well previous years' exam questions fit to this year's course. Questions fall into the following categories:

- A **good fit** means an exam question could look like this.
- A **reasonable fit** means the mathematical techniques are relevant but the style is different. A such, it is worth doing if you have exhausted all the 'good fit' questions.
- If a question is not listed it is **not relevant** to this year's course. If specific parts aren't mentioned, then all of the question is relevant.

Terminology changes a bit throughout the papers but know that

- Critical point = steady state = fixed point = equilibrium.
- Constant non-zero steady state = spatially homogeneous equilibrium = homogeneous equilibrium.
- Physically meaningful equilibrium = permissible equilibrium = feasible equilibrium.
- Classify the equilibrium... = what type of equilibrium... = decide whether it is a stable node etc.

2025 paper

Q1 Good fit.

Q2 Reasonable fit.

Q3 Good fit.

Q4 Good fit.

Q5 Good fit.

Q6 Reasonable fit.

Q7 Good fit.

Q8 Good fit.

2024 paper

Q1 Good fit.

Q2 Good fit.

Q3 Good fit.

Q4 Good fit.

Q5 Reasonable fit.

Q6 (a)–(c) Good fit. (d) is subtly different to the cases we've seen in class because there is no growth term, which means all u are homogeneous equilibria and the boundary conditions are different.

Q7 Good fit.

Q8 Good fit.

2023 paper

Q1 Good fit.

Q2 Good fit.

Q3 Good fit.

Q4 Good fit.

Q5 Good fit.

Q6 Good fit.

Q7 Good fit.

Q8 Good fit.

2022 paper

Q1 Good fit.

Q2 Good fit.

Q3 Good fit.

Q4 Good fit.

Q5 Good fit.

Q6 Good fit.

Q7 Good fit.

Q8 Good fit.

2021 paper

Take-home paper, so questions are longer than you would expect this year.

Q1 Good fit.

Q2 Good fit.

Q3 Good fit.

Q4 Good fit.

2020 paper

Take-home paper, so questions are longer than you would expect this year.

Q1 Good fit.

Q3 Good fit.

Q4 Good fit.

Q5 Good fit.

2019 paper

Q1 Good fit.

Q2 Good fit.

Q3 Good fit.

Q4 Good fit.

Q5 Reasonable fit.

Q6 Good fit.

Q7 Good fit.

Q8 Good fit.

Q9 Good fit.

2018 paper

Q1 Good fit.

Q2 Good fit.

Q3 Good fit.

Q4 Good fit.

Q5 Good fit.

Q6 Good fit.

Q7 Good fit.

Q8 Good fit.

Q10 Reasonable fit.

2017 paper

- Q1 Good fit.
- Q2 Good fit.
- Q3 Reasonable fit.
- Q4 Good fit.
- Q5 Good fit.
- Q6 Good fit.
- Q7 Good fit.
- Q8 Good fit.

2016 paper

- Q1 Good fit.
- Q2 Good fit.
- Q3 Good fit.
- Q5 Good fit.
- Q6 Good fit.
- Q7 Good fit.
- Q8 Good fit. In (e), conditions for aggregation mean satisfying the Turing conditions.
- Q9 Good fit.
- Q10 Reasonable fit. In (d), the linear expansion is $u = u_0 + \delta u$.

2015 paper

- Q1 Reasonable fit.
- Q3 (a)–(b) Good fit.
- Q4 Good fit. Linear (instability) analysis = linear stability analysis.
- Q6 Good fit.
- Q7 (a)–(e) Reasonable fit.
- Q8 Good fit.

- Q9 Reasonable fit.

- Q10 Good fit, but they would have seen this model already in class.

2014 paper

- Q1 Good fit. The definition of J is such that $\partial\rho/\partial t = -\partial J/\partial x$. You would be given J in an exam.
- Q3 Good fit.
- Q4 Good fit.
- Q5 Good fit.
- Q6 Good fit.
- Q7 Good fit. In (c), the linear expansion is $u = u^* + v$ where v is small (so these are the $\mathcal{O}(v)$ equations).
- Q8 Apart from (d), reasonable fit.
- Q10 Good fit. In (d), conditions for aggregation mean satisfying the Turing conditions.

2013 paper

- Q1 (a), (b) Good fit. (c), (d) Reasonable fit.
- Q2 Good fit.
- Q4 Good fit.
- Q5 Good fit.
- Q6 Reasonable fit.
- Q7 Good fit. The definition of J is such that $\partial c/\partial t = -\partial J/\partial x$ and the mean transit time is N/J .
- Q8 Reasonable fit.
- Q9 Good fit.

Q10 Reasonable fit. Note that $\Delta = \nabla^2$, the Laplacian. In (d) the perturbation is in the form $\rho = \rho_0 + \delta\rho$ where $\delta\rho$ is small. In (f) aggregation implies the Turing instability. Aggregation comes from requiring the instability by $\sigma > 0$.

2012 paper

Q2 (a)–(c) Good fit.

Q3 (a), (b) Reasonable fit.

Q4 Good fit.

Q5 Good fit.

Q8 Good fit, but they would have seen this model already in class.

Q9 Reasonable fit.

Q10 Reasonable fit.

2011 paper

Q1 Good fit

Q2 Reasonable fit. Not sure if this is how Shakespeare meant it, mind you.

Q3 Good fit.

Q4 Reasonable fit.

Q5 Good fit.

Q6 Good fit. For (d) two possibilities should be enough. Chemotactic wave = travelling wave.

Q7 Good fit. Use Fourier transforms to solve this directly (I think here they are looking for you to show it is true by differentiation.)

Q8 Reasonable fit.

Q9 Reasonable fit.

Q10 Good fit.